

Technology Stack

Date	14 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit (Python), HTML, CSS	Provides an interactive, responsive, and easy-to-use web interface for users to enter text and view emotion predictions with personalized learning recommendations.

2	Backend / Server-Side	Python 3.12, Streamlit, TensorFlow, PyTorch, Hugging Face Transformers	Handles text preprocessing, emotion prediction, model inference, and AI response generation efficiently using modern machine learning frameworks.
3	Database / Data Storage	CSV Files, JSON Configuration Files, Pickle (.pkl), Safetensors (.safetensors), Keras (.keras)	Stores prediction history, label mappings, tokenizer configurations, trained models, and application settings without requiring a traditional database.
4	Cloud / Hosting / Deployment	Streamlit Local Deployment, GitHub Repository	Enables easy deployment, source code management, project sharing, and future cloud deployment possibilities.

5	Version Control & CI/CD	Git, GitHub, Git LFS	Provides version control, collaborative development, secure storage of large machine learning models, and project tracking.
6	Third-Party APIs / Other Tools	Google Gemini API, Hugging Face Transformers, NumPy, Pandas, Scikit-learn	Gemini API generates personalized learning guidance. Hugging Face supports BERT inference. NumPy, Pandas, and Scikit-learn are used for preprocessing, data handling, and machine learning operations.